

PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON EXPERIENCE (IOHE)

ON

Social Media Analytics for Trend Prediction Using Machine Learning

submitted in partial fulfilment of the requirements for the award of degree of

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In

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled “**Social Media Analytics for Trend Prediction Using Machine Learning**” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Ms. Swati Goel**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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ABSTRACT

The exponential rise of social media has created an unprecedented stream of user-generated content, encompassing posts, hashtags, comments, and shares that collectively reflect evolving public sentiment and emerging trends in near real-time. Traditional trend analysis methodologies, which rely on periodic surveys and retrospective historical data, are inherently slow and fail to capture the dynamic nature of online discourse. This paper presents a comprehensive machine learning (ML) framework for social media trend prediction, integrating Natural Language Processing (NLP) pre-processing techniques with multiple classification algorithms. The proposed system employs TF-IDF feature extraction combined with hashtag frequency analysis and engagement-based metrics to construct a rich feature space. Three supervised learning models-Logistic Regression, Naive Bayes, and Random Forest are systematically trained, validated, and compared. Experimental results demonstrate that the Random Forest classifier achieves the highest trend prediction accuracy of 91%, outperforming competing models across all evaluation metrics. Time series visualisations further validate the system's capability to detect trend onset approximately 4.2 hours before official trending

Keywords: Social Media Analytics, Trend Prediction, Machine Learning, Natural Language Processing, TF-IDF, Hashtag Analysis, Random Forest, Sentiment Analysis.

1. Introduction

1.1. Background

The modern digital ecosystem generates over 500 million social media posts per day. This real-time content stream mirrors public interests, political discourse, and cultural phenomena. Traditional approaches surveys and historical data analysis are slow and non-adaptive. Machine Learning (ML) combined with Natural Language Processing (NLP) enables automated, real-time discovery of trending patterns

1.2. Problem Statement

Existing trend detection systems are reactive they identify trends only after they have already surfaced on platform dashboards. There is a clear need for a proactive, predictive system capable of identifying emerging trends before they officially appear, enabling timely response by marketers, journalists, and crisis communicators

1.3. Objectives

- To design and implement an end-to-end ML pipeline for social media trend prediction.
- To perform a systematic comparative study of three supervised ML. classifiers: Logistic Regression, Naive Bayes, and Random Forest.
- To evaluate the predictive accuracy, precision, recall, and F1-score of each model.
- To identify the most discriminative features that drive trend emergence.
- To validate the system's early-warning capability using time-series analysis.

1.4. Scope

This project is scoped to English language Twitter data spanning six months. The dataset covers five domains: technology, entertainment, politics, sports, and public health. The system is designed for offline batch processing with future potential for real-time integration.

2. Methodology

2.1.Dataset

A corpus of approximately 150,000 Twitter posts was collected via the Academic Research API over a period of six months. The dataset spans five domains: technology, entertainment, politics, sports, and public health. Each record includes timestamp, text content, hashtags, like count, retweet count, and a binary trending label. The dataset exhibits class imbalance (31% trending, 69% non-trending), which was addressed through stratified splits and class-weight adjustment during model training

2.2.Preprocessing Pipeline

Raw text data underwent a multi-stage preprocessing pipeline to standardise vocabulary and reduce noise

- URL and mention removal
- Lowercase normalisation
- Punctuation stripping
- Whitespace tokenisation
- NLTK stop-word removal
- WordNet Lemmatization

2.3.Feature Engineering

2.3.1.TF-IDF Textual Features

Preprocessed text was vectorised using TF-IDF with a maximum vocabulary of 10,000 unigrams and bigrams. TF-IDF weights each term by its frequency in the document multiplied by the inverse document frequency, amplifying domain specific discriminative vocabulary.

2.3.2.Hashtag Velocity Features

Per post features include unique hashtag count, 24-hour usage frequency, and frequency rate-of-change (velocity). A sudden velocity spike is a strong indicator of imminent trending behaviour.

2.3.3.Engagement Metrics

Min-max normalised like count, retweet count, and retweet to like ratio capture network amplification behaviour linked to algorithmic trending thresholds.

2.4.Machine Learning Models

Three supervised learning models were selected and tuned via 5-fold cross-validated grid search:

- Logistic Regression (L2 regularised) - a linear probabilistic classifier.
- Multinomial Naive Bayes - a probabilistic model based on Bayes' theorem suited for text classification.
- Random Forestan ensemble of decision trees using bagging for robust classification.

2.5.Evaluation Protocol

An 80/20 stratified train-test split was employed. Performance was reported on the held-out test set using Accuracy, Precision, Recall, and F1-Score. All experiments were implemented in Python 3.10 with Scikit-learn 1.3, Pandas, and NLTK.

3. Tools and Technologies

3.1. Programming Language

Python 3.10-chosen for its extensive ML and NLP ecosystem.

3.2. Machine Learning Libraries

- Scikit-learn 1.3-for model training, evaluation, and grid search.
- NumPy and Pandas for data manipulation and numerical computation.

3.3. Natural Language Processing

- NLTK (Natural Language Toolkit)- for tokenisation, stop-word removal, and lemmatisation.
- VADER Sentiment Analyser for rule-based sentiment polarity scoring of social media text.

3.4. Data Collection

- Twitter Academic Research API for dataset collection.
- Tweepy - Python wrapper for the Twitter API.

3.5. Visualisation

- Matplotlib for accuracy comparison and feature importance bar charts.
- Seaborn for grouped precision-recall-F1 bar plots.

3.6. Development Environment

- Jupyter Notebook- for iterative data analysis and model development.
- VS Code for scripted pipeline components
- Git/GitHub-for version control

4. IMPLEMENTATION

4.1. System Architecture

The proposed system follows a multi-stage implementation pipeline:

- Data Collection
- Data Preprocessing
- Feature Extraction
- Model Training
- Model Evaluation
- Trend Prediction

The pipeline begins with data collection from social media platforms using APIs. The collected textual data is then cleaned and transformed using NLP preprocessing techniques.

Feature engineering methods are applied to generate meaningful numerical representations from textual and engagement data. The processed features are then used to train multiple machine learning models.

Finally, the trained models are evaluated on unseen test data to determine predictive performance.

4.2. Data Preprocessing Implementation

Python libraries such as Pandas and NLTK were used to preprocess the data. The preprocessing pipeline involved:

- Removing hyperlinks and mentions
- Converting text to lowercase
- Removing punctuation marks
- Tokenization
- Stop-word filtering
- Lemmatization

The preprocessing stage significantly improved data quality and reduced dimensionality.

4.3.Feature Extraction Implementation

TF-IDF vectorization was implemented using Scikit-learn's TfidfVectorizer module. Additional numerical features related to hashtags and engagement were integrated with the textual feature vectors.

The combined feature set provided richer contextual information for machine learning models.

4.4.Model Training

The dataset was divided into training and testing sets using stratified sampling. Three machine learning classifiers were trained:

- Logistic Regression
- Naive Bayes
- Random Forest

Cross-validation and grid search were applied to optimize model parameters and improve prediction accuracy.

4.5.Visualization and Analysis

Graphs and charts were generated using Matplotlib and Seaborn to compare:

- Model accuracy
- Precision, Recall, and F1-score
- Feature importance
- Cross-validation performance
- Time-series trend prediction

The visual analysis helped identify the strengths and limitations of each model.

5. MAJOR FINDINGS / OUTCOMES / RESULTS

5.1. Model Performance Comparison

The experimental results demonstrated that the Random Forest classifier achieved the highest performance among all models.

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	82.3%	80.1%	78.6%	79.3%
Naive Bayes	76.8%	74.5%	73.2%	73.8%
Random Forest	91.0%	89.5%	88.7%	89.1%

5.2. Feature Importance Analysis

Feature importance analysis revealed that the most influential predictors were:

- Hashtag velocity
- Retweet count
- Sentiment polarity
- Engagement ratio
- Trending keywords

The analysis showed that engagement-based features significantly contribute to trend prediction.

5.3. Early Trend Detection

The system successfully predicted trend emergence approximately 4.2 hours before official trend onset on the platform. This demonstrates the effectiveness of machine learning techniques for proactive trend detection.

5.4. Cross-Validation Stability

The Random Forest classifier demonstrated stable performance across all folds during cross-validation testing, achieving a mean accuracy of approximately 91.0% with low

variance.

6. CONCLUSION AND FUTURE SCOPE

6.1.Conclusion

This project presented a complete machine learning-based framework for social media trend prediction using Natural Language Processing and supervised learning techniques.

The system successfully integrated textual features, hashtag dynamics, and engagement metrics to identify emerging trends on social media platforms. Among the evaluated classifiers, Random Forest achieved the best predictive performance with an accuracy of 91%.

The experimental results demonstrated that machine learning techniques can effectively support proactive trend prediction and early detection systems. The project also highlighted the importance of feature engineering and engagement analysis in improving predictive accuracy.

The developed framework has practical applications in digital marketing, public opinion monitoring, journalism, healthcare communication, and crisis management.

6.2.Future Scope

The project can be further enhanced through:

- Real-time streaming analytics using Apache Kafka and Spark Streaming
- Transformer-based deep learning models such as BERT and BERTweet
- Multilingual trend prediction support
- Cross-platform analytics integrating Twitter, Instagram, Reddit, and Facebook
- Improved bot detection mechanisms
- Graph-based propagation analysis for viral content prediction

Future enhancements can significantly improve scalability, robustness, and prediction accuracy.

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APPENDICES

Appendix A: Sample Python Libraries Used

- pandas
- numpy
- scikit-learn
- nltk
- matplotlib
- seaborn
- tweepy

Appendix B: Sample Workflow

- Collect Twitter Data
- Preprocess Text
- Extract Features
- Train Machine Learning Models
- Evaluate Performance
- Predict Emerging Trends

Appendix C: Expected Outputs

- Trend prediction probabilities
- Comparative performance graphs
- Feature importance charts
- Cross-validation reports
- Trend onset visualizations